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WIRE DAMPER OF LOUDSPEAKER WITH PERIODIC SPAN WIRES AND THE METHOD FOR MANUFACTURING THE SAME

Abstract

A method for manufacturing a wire damper of a loudspeaker with periodic span wires, including: arranging warp yarns and wire assemblies at intervals, which extend straightly and are parallel to each other; interweaving weft yarns with the warp yarns and the wire assemblies to weave a base material, each wire assembly periodically spans above and below the weft yarn assemblies, and an area where the weft yarn assemblies defined by at least two adjacent weft yarns are interwoven with each wire assembly is defined as a wire disposing area; impregnating the base material in a resin solution; drying the base material; thermoforming a wire damper on the base material; separating the wire damper from the base material. Therefore, the strength of the weft yarns fixing the wire assemblies can be reduced, thereby reducing the influence of the weft yarns on the wire assemblies, imparting the wire assemblies better elasticity.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority of Taiwanese patent application No. 113108004, filed on Mar. 5, 2024, which is incorporated herewith by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a wire damper of a loudspeaker and the method for manufacturing the same, and more particularly, to a wire damper of a loudspeaker with periodic span wires and the method for manufacturing the same.

2. The Prior Arts

[0003] In the general moving coil loudspeaker, the principle that the reaction force of a fixed magnetic field causes another magnetic field to move in the opposite direction (i.e., opposite magnetisms attract each other, and like magnetisms repels each other) is used to produce sound. Further, the power alternating current generated by the power amplifier is transmitted to the voice coil through a wire to change the polarity of the magnetic field, such that the voice coil generates a reaction force against the fixed magnetic region generated by the magnetic circuit device. A positive pulse causes an outward motion of the diaphragm relative to the magnet, while a negative pulse causes an inward motion of the diaphragm. When the voice coil pushes the diaphragm to reciprocate, the diaphragm pushes air, and the air pressure changes to form sound waves. The damper is responsible for maintaining the voice coil at a correct position in the gap of the magnet core, ensuring that the voice coil reciprocates along the axis direction when being forced.

[0004] A conventional wire damper of a loudspeaker comprises a main body, four wire assemblies and a resin solid layer. The main body includes a plurality of warp yarns and a plurality of weft yarns, and the weft yarns are interwoven with the warp yarns and the wire assemblies. Each wire assembly is composed of a plurality of wires. The weft yarns jointly form two elastic adjustment areas between the warp yarns closest to both sides of the wire assemblies and the wire assemblies, the widths of the elastic adjustment areas are equal to each other and greater than the distance between the warp yarns. The solid resin layer covers surfaces of the warp yarns, the weft yarns and the wires.

[0005] Because the wire assemblies are harder than the warp yarns and the weft yarns, and the elasticity and toughness of the wire assemblies are worse than those of the warp yarns and the weft yarns, in the conventional wire damper of the loudspeaker, the hardness, the elasticity and the toughness of the wire disposing areas can be jointly adjusted by disposing the elastic adjustment areas.

[0006] However, since two adjacent weft yarns pass above and below the wire assemblies respectively, although the weft yarns can firmly fix the wire assemblies through weaving, the fixing strength applied by the weft yarns is quite great, which affects the elasticity of the wire assemblies, causing the effect of jointly adjusting the hardness, the elasticity and the toughness of the wire disposing areas by the elastic adjustment areas to be less effective than expected. The wire disposing areas are still harder than the other areas of the wire damper of the loudspeaker, and the elasticity and the toughness of the wire disposing areas are still worse than those of the other areas of the wire damper of the loudspeaker. Therefore, the hardness, the elasticity and the toughness of the wire damper of the loudspeaker are not uniform, causing the elastic resilience and the fatigue resistance of the wire damper of the loudspeaker to be not uniform. Therefore, the wire damper of the loudspeaker is prone to be deformed, thereby affecting the output sound quality of the

loudspeaker.

SUMMARY OF THE INVENTION

[0007] A main objective of the present invention is to provide a wire damper of a loudspeaker with periodic span wires and the method for manufacturing the same, which can reduce the strength of the weft yarns fixing the wire assemblies.

[0008] In order to achieve the above objective, the present invention provides a method for manufacturing a wire damper of a loudspeaker with periodic span wires, comprising the following steps: (a) arranging a plurality of warp yarns and a plurality of wire assemblies at intervals in a manner that the warp yarns and the wire assemblies extend straightly along a first direction and are parallel to each other, wherein each wire assembly is composed of a plurality of wires, and each wire is a monofilament thread; (b) interweaving a plurality of weft yarns with the warp yarns and the wire assemblies along a second direction to weave a base material, wherein the first direction is perpendicular to the second direction, at least two adjacent weft yarns are defined as a weft yarn assembly, each wire assembly periodically spans above and below the weft yarn assemblies along the first direction, and an area where the weft yarn assemblies are interwoven with each wire assembly is defined as a wire disposing area; (c) impregnating the base material in a resin solution; (d) drying the base material to form a solid resin layer on the base material; (e) forming a wire damper of a loudspeaker on the base material by thermoforming; and (f) separating the wire damper of the loudspeaker from the base material.

[0009] In some embodiments, in the step (a), the wires are arranged at intervals from each other.

[0010] In some embodiments, in the step (a), the wires are blend-twisted such that each wire assembly forms a multifilament thread with a circular cross-section.

[0011] In some embodiments, in the step (a), the wires are braided with each other such that each wire assembly forms a multifilament thread with a flat cross-section.

[0012] In some embodiments, in the step (e), the weft yarns jointly form a first elastic adjustment area between a first side of each wire assembly and the warp yarn closest to the first side of each wire assembly, the weft yarns jointly form a second elastic adjustment area between a second side of each wire assembly and the warp yarn closest to the second side of each wire assembly, a width of the first elastic adjustment area and a width of the second elastic adjustment area are equal to each other, a distance between the warp yarns is smaller than the width of the first elastic adjustment area and the width of the second elastic adjustment area.

[0013] In order to achieve the above objective, the present invention provides a wire damper of a loudspeaker with periodic span wires, comprising a main body, a plurality of wire assemblies and a solid resin layer. The main body includes a plurality of warp yarns and a plurality of weft yarns, the warp yarns are arranged at intervals and extend straightly along a first direction and are parallel to each other, the weft yarns are interwoven with the warp yarns along a second direction, the first direction is perpendicular to the second direction, and at least two adjacent weft yarns are defined as a weft yarn assembly. A plurality of wire assemblies are arranged at intervals with the warp yarns, extend straightly along the first direction, are parallel to the warp yarns, and cross above and below the weft yarn assemblies along the first direction periodically, wherein each wire assembly is composed of a plurality of wires, each wire is a monofilament thread, and an area where the weft yarn assemblies are interwoven with each wire assembly is defined as a wire disposing area. The solid resin layer covers surfaces of the warp yarns, the weft yarns and the wires.

[0014] In some embodiments, the wires are arranged at intervals from each other.

[0015] In some embodiments, the wires are blend-twisted such that each wire assembly forms a multifilament thread with a circular cross-section.

[0016] In some embodiments, the wires are braided with each other such that each wire assembly forms a multifilament thread with a flat cross-section.

[0017] In some embodiments, the weft yarns jointly form a first elastic adjustment area between a first side of each wire assembly and the warp yarn closest to the first side of each wire assembly,

the weft yarns jointly form a second elastic adjustment area between a second side of each wire assembly and the warp yarn closest to the second side of each wire assembly, a width of the first elastic adjustment area and a width of the second elastic adjustment area are equal to each other, a distance between the warp yarns is smaller than the width of the first elastic adjustment area and the width of the second elastic adjustment area.

[0018] The effect of the present invention is in that by disposing the wire assemblies in a periodically spanning manner, the strength of the weft yarns fixing the wire assemblies can be reduced, thereby reducing the influence of the weft yarns on the wire assemblies, so as to impart the wire assemblies better elasticity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a flow diagram of a manufacturing method of the present invention;

[0020] FIG. 2 is a schematic diagram of a step S10 and a step S20 of the manufacturing method of a first embodiment of the present invention;

[0021] FIG. 3 is a schematic diagram of step S10 to step S50 of the manufacturing method of the first embodiment of the present invention;

[0022] FIG. 4 is a perspective view of a wire damper of a loudspeaker of the first embodiment of the present invention;

[0023] FIG. 5 is a schematic diagram of a region A of FIG. 4;

[0024] FIG. 6 is a schematic diagram of a step S10 and a step S20 of the manufacturing method of a second embodiment of the present invention;

[0025] FIG. 7 is a perspective view of a wire damper of a loudspeaker of the second embodiment of the present invention;

[0026] FIG. 8 is a schematic diagram of a region B of FIG. 7;

[0027] FIG. 9 is a schematic diagram of a step S10 and a step S20 of the manufacturing method of a third embodiment of the present invention;

[0028] FIG. 10 is a perspective view of a wire damper of loudspeaker of the third embodiment of the present invention; and

[0029] FIG. 11 is a schematic diagram of a region C of FIG. 7.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0030] Hereinafter, embodiments of the present invention will be described in more detail with reference to the drawings and the reference numerals, such that those skilled in the art can implement it after studying this specification.

[0031] FIG. 1 is a flow diagram of a manufacturing method of the present invention. FIG. 2 is a schematic diagram of a step S10 and a step S20 of the manufacturing method of a first embodiment of the present invention. FIG. 3 is a schematic diagram of step S10 to step S50 of the manufacturing method of the first embodiment of the present invention. The present invention provides a method for manufacturing a wire damper of a loudspeaker with periodic span wires, comprising the following steps S10 to S60:

[0032] In the step S10, as shown in FIGS. 1 and 2, a plurality of warp yarns 10 and a plurality of wire assemblies 20 are arranged at intervals in a manner that the warp yarns 10 and the wire assemblies 20 extend straightly along a first direction and are parallel to each other. Wherein each wire assembly 20 is composed of a plurality of wires 21, each wire 21 is a monofilament thread, and the wires 21 are arranged at intervals. In other words, the warp yarns 10 and the wires 21 are spaced from each other at a same interval.

[0033] In the step S20, as shown in FIGS. 1 and 2, a plurality of weft yarns 31 are interwoven with the warp yarns 10 and the wire assemblies 20 along a second direction to weave a base material 40.

Wherein the first direction is perpendicular to the second direction, and three adjacent weft yarns **31** are defined as a weft yarn assembly **30**, each wire assembly **20** spans above and below the weft yarn assemblies **30** along the first direction periodically, and an area where the weft yarn assemblies **30** are interwoven with wire assemblies **20** is defined as a wire disposing area **41**. To be more specific, the term “periodically spanning” means that each wire assembly **20** continuously and alternately spans the weft yarn assemblies **30**. For example, firstly, each wire assembly **20** spans above three weft yarns **31** of one of the weft yarn assemblies **30**, and then each wire assembly **20** spans below three weft yarns **31** of another weft yarn assembly **30**, and so on.

[0034] In some embodiments, the number of the weft yarns **31** in each weft yarn assembly is at least two. First, each wire assembly **20** spans above the two weft yarns **31** of one of the weft yarn assembly **30**, and then each wire assembly **20** spans below the two weft yarns **31** of the another weft yarn assembly **30**.

[0035] In some embodiments, the number of weft yarns **31** of the weft yarn assembly **30** may be different. For example, the number of weft yarns **31** of some weft yarn assembly **30** are two, while in other warp assembly **30**, the number of warp yarns **31** is three. First, each wire assembly **20** spans above the two weft yarns **31** of one of the weft yarn assembly **30**, and then each wire assembly **20** spans below the three weft yarns **31** of another weft yarn assembly **30**. Next, each wire assembly **20** spans above the two weft yarns **31** of another weft yarn assembly **30**, and then each wire assembly **20** spans below the three weft yarns **31** of another weft yarn assembly **30**, and so on.

[0036] In the step S30, as shown in FIGS. 1 and 3, a base material **40** is impregnated in a resin solution **51** in a resin tank **50**, such that the warp yarns **10**, the wires **21** and the weft yarns **31** adsorb the resin and are adhered with the resin.

[0037] In the step S40, as shown in FIGS. 1 and 3, a drying device **60** includes an upper baking plate **61** and a lower baking plate **62**. By utilizing the temperature of the upper baking plate **61** and the lower baking plate **62**, the moisture and volatile substances in the resin on the base material **40** are removed such that the base material **40** is dried. Meanwhile, the resin penetrates into the base material **40** and is adhered onto the warp yarns **10**, the wires **21** and the weft yarns **31**, so as to form a solid resin layer **130** (referring to FIG. 5).

[0038] In the step S50, as shown in FIGS. 1 and 3, a thermoforming device **70** includes a forming mold **71** and a heating device (not shown), and the forming mold **71** includes an upper mold **711** and a lower mold **712**. When the upper mold **711** and the lower mold **712** fit together and pressurize the base material **40**, the heating device is applied with electricity to increase the temperature of the upper mold **711** and the lower mold **712** to a range of 190 to 270° C., thereby softening the resin on the base material **40**. In addition to destroying the resin structure, the resin also fills up in the gaps, and thus respective parts of the resin are connected with each other to form the final morphology of the solid resin layer **130**. Therefore, the resin cover between the warp yarns **10**, the wires **21** and the weft yarns **31**, thereby forming a wire damper of loudspeaker **100** on the base material **40** by thermoforming. The weft yarns **31** jointly form a first elastic adjustment area **101** between a first side of the wire assemblies **20** and the warp yarn **10** closest to the first side of the wire assemblies **20** (referring to FIG. 5), the weft yarns **31** jointly form a second elastic adjustment area **102** between a second side of the wire assemblies **20** and the warp yarns **10** closest to the second side of the wire assemblies **20** (referring to FIG. 5), a width of the first elastic adjustment area **101** and a width of the second elastic adjustment area **102** are equal to each other, a distance between the warp yarns is smaller than the width of the first elastic adjustment area and the width of the second elastic adjustment area.

[0039] In the step S60, as shown in FIGS. 1 and 3, a cutting device **80** includes an upper cutting tool **81** and a lower cutting tool **82**, wherein the wire damper of loudspeaker **100** is cut from the base material **40** by the upper cutting tool **81** and the lower cutting tool **82**, such that the wire damper of loudspeaker **100** is separated from the base material **40**.

[0040] FIG. 4 is a perspective view of a wire damper of loudspeaker 100 of the first embodiment of the present invention. FIG. 5 is a schematic diagram of a region A of FIG. 4. As shown in FIGS. 4 and 5, the present invention provides a wire damper of loudspeaker 100 with periodic span wires comprises a base material 30, a wire damper of loudspeaker 110, a plurality of wire assemblies 20 and a solid resin layer 130.

[0041] The main body 110 includes a plurality of warp yarns 10 and a plurality of weft yarns 31, the warp yarns are arranged at intervals and extend straightly along a first direction and are parallel to each other, the weft yarns 31 are interwoven with the warp yarns 10 along a second direction, the first direction is perpendicular to the second direction, and adjacent three weft yarns 31 are defined as a weft yarn assembly 30.

[0042] In some embodiments, the number of the weft yarns 31 in each weft yarn assembly is at least two.

[0043] In some embodiments, the numbers of the weft yarns 31 of the weft yarn assemblies 30 may be different. For example, the numbers of the weft yarns 31 of some weft yarn assemblies 30 are two, while the number of the warp yarns 31 of some warp assemblies 30 are three.

[0044] The wire assemblies 20 are arranged at intervals with the warp yarns 10, extend straightly along the first direction, are parallel to the warp yarns 10, and periodically span above and below the weft yarn assemblies 30 along the first direction. More explicitly, the term “periodical span” refers to continuously and alternately spanning. For example, each wire assembly 20 first spans above one of the weft yarn assembly 30, and then spans below another weft yarn assembly 30, and so on. Wherein each wire assembly 20 is composed of a plurality of wires 21, each wire 21 is a monofilament thread, and the wires 21 are arranged at intervals. Areas where the weft yarn assemblies 30 are interwoven with the wire assemblies 20 are defined as two wire disposing areas 41. The weft yarns 31 jointly form a first elastic adjustment area 101 between a first side of the wire assemblies 20 and the warp yarn 10 closest to the first side of the wire assemblies 20, the weft yarns 31 jointly form a second elastic adjustment area 102 between a second side of the wire assemblies 20 and the warp yarns 10 closest to the second side of the wire assemblies 20, a width of the first elastic adjustment area 101 and a width of the second elastic adjustment area 102 are equal to each other, a distance between the warp yarns 10 is smaller than the width of the first elastic adjustment area 101 and the width of the second elastic adjustment area 102.

[0045] The solid resin layer 130 covers surfaces of the warp yarns 10, the weft yarns 31 and the wires 21.

[0046] FIG. 6 is a schematic diagram of a step S10 and a step S20 of the manufacturing method of a second embodiment of the present invention. As shown in FIG. 6, in terms of method, the second embodiment is different from the first embodiment in that in the step S10, the wires 21 are blend-twisted, such that each wire assembly 20A forms a multifilament thread with a circular cross-section.

[0047] FIG. 7 is a perspective view of a wire damper of loudspeaker 100A of the second embodiment of the present invention. FIG. 8 is a schematic diagram of a region B of FIG. 7. As shown in FIGS. 7 and 8, in terms of structure, the wires 21 are blend-twisted, such that each wire assembly 20A forms a multifilament thread with a circular cross-section.

[0048] FIG. 9 is a schematic diagram of a step S10 and a step S20 of the manufacturing method of a third embodiment of the present invention. As shown in FIG. 9, in terms of method, the third embodiment is different from the previous embodiments in that in the step S10, the wires 21 are braided with each other, such that each wire assembly 20B forms a multifilament thread with a flat cross-section.

[0049] FIG. 10 is a perspective view of a wire damper of loudspeaker 100B of the third embodiment of the present invention. FIG. 11 is a schematic diagram of a region C of FIG. 10. As shown in FIGS. 10 and 11, in terms of structure, the wires 21 are braided with each other, such that each wire assembly 20B forms a multifilament thread with a flat cross-section.

[0050] In summary, in the present invention, by disposing the wire assemblies **20**, **20A** and **20B** in a periodically spanning manner, the strength of the weft yarns **31** fixing the wire assemblies **20**, **20A** and **20B** can be reduced, thereby reducing the influence of the weft yarns **31** on the wire assemblies **20**, **20A** and **20B**, so as to impart the wire assemblies **20**, **20A** and **20B** better elasticity. [0051] It is noted that in the present invention, by disposing the first elastic adjustment area **101** and the second elastic adjustment area **102** in a periodically spanning manner, as well as jointly adjusting the hardness, the elasticity and the toughness of the wire disposing areas **41** by the first elastic adjustment area **101** and the second elastic adjustment area **102**, the wire disposing areas **41** become softer, and the elasticity and the toughness of the wire disposing areas **41** are improved, such that the hardness, the elasticity and the toughness of the wire disposing areas **41** and the wire assemblies **20**, **20A** and **20B** are comparable to those of the other areas of the wire dampers of loudspeaker **100**, **100A** and **100B**. Therefore, the wire dampers of loudspeaker **100**, **100A** and **100B** have uniform hardness, elasticity and toughness, thereby having uniform elastic resilience and fatigue resistance, are not easily deformed and brittle, and have improved output sound quality of the loudspeaker.

[0052] Those mentioned above are only preferred embodiments for explaining the present invention, but intend to limit the present invention in any forms. Therefore, any modifications or changes related to the present invention made in the same invention spirit should still be included in the scope of the present invention as intended to be claimed.

Claims

1. A method for manufacturing a wire damper of a loudspeaker with periodic span wires, comprising the following steps: (a) arranging a plurality of warp yarns and a plurality of wire assemblies at intervals in a manner that the warp yarns and the wire assemblies extend straightly along a first direction and are parallel to each other, wherein each wire assembly is composed of a plurality of wires, and each wire is a monofilament thread; (b) interweaving a plurality of weft yarns with the warp yarns and the wire assemblies along a second direction to weave a base material, wherein the first direction is perpendicular to the second direction, at least two adjacent weft yarns are defined as a weft yarn assembly, each wire assembly periodically spans above and below the weft yarn assemblies along the first direction, and an area where the weft yarn assemblies are interwoven with each wire assembly is defined as a wire disposing area; (c) impregnating the base material in a resin solution; (d) drying the base material to form a solid resin layer on the base material; (e) forming a wire damper of a loudspeaker on the base material by thermoforming; and (f) separating the wire damper of the loudspeaker from the base material.
2. The method according to claim 1, wherein in the step (a), the wires are arranged at intervals from each other.
3. The method according to claim 1, wherein in the step (a), the wires are blend-twisted such that each wire assembly forms a multifilament thread with a circular cross-section.
4. The method according to claim 1, wherein in the step (a), the wires are braided with each other such that each wire assembly forms a multifilament thread with a flat cross-section.
5. The method according to claim 1, wherein in the step (e), the weft yarns jointly form a first elastic adjustment area between a first side of each wire assembly and the warp yarn closest to the first side of each wire assembly, the weft yarns jointly form a second elastic adjustment area between a second side of each wire assembly and the warp yarn closest to the second side of each wire assembly, a width of the first elastic adjustment area and a width of the second elastic adjustment area are equal to each other, a distance between the warp yarns is smaller than the width of the first elastic adjustment area and the width of the second elastic adjustment area.
6. A wire damper of a loudspeaker with periodic span wires, comprising: a main body, which includes a plurality of warp yarns and a plurality of weft yarns, the warp yarns are arranged at

intervals and extend straightly along a first direction and are parallel to each other, the weft yarns are interwoven with the warp yarns along a second direction, the first direction is perpendicular to the second direction, and at least two adjacent weft yarns are defined as a weft yarn assembly; a plurality of wire assemblies, which are arranged at intervals with the warp yarns, extend straightly along the first direction, are parallel to the warp yarns, and periodically span above and below the weft yarn assemblies along the first direction, wherein each wire assembly is composed of a plurality of wires, each wire is a monofilament thread, and an area where the weft yarn assemblies are interwoven with each wire assembly is defined as a wire disposing area; and a solid resin layer, which covers surfaces of the warp yarns, the weft yarns and the wires.

7. The wire damper of the loudspeaker according to claim 6, wherein the wires are arranged at intervals from each other.

8. The wire damper of the loudspeaker according to claim 6, wherein the wires are blend-twisted such that each wire assembly forms a multifilament thread with a circular cross-section.

9. The wire damper of the loudspeaker according to claim 6, wherein the wires are braided with each other such that each wire assembly forms a multifilament thread with a flat cross-section.

10. The wire damper of the loudspeaker according to claim 6, wherein the weft yarns jointly form a first elastic adjustment area between a first side of each wire assembly and the warp yarn closest to the first side of each wire assembly, the weft yarns jointly form a second elastic adjustment area between a second side of each wire assembly and the warp yarn closest to the second side of each wire assembly, a width of the first elastic adjustment area and a width of the second elastic adjustment area are equal to each other, a distance between the warp yarns is smaller than the width of the first elastic adjustment area and the width of the second elastic adjustment area.
